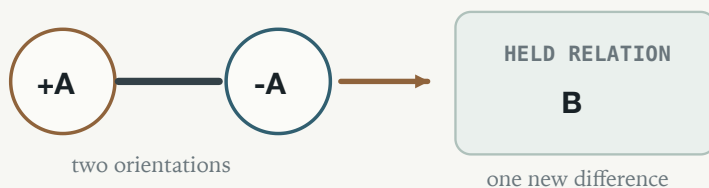


A SHORT AXIOMATIC ACCOUNT

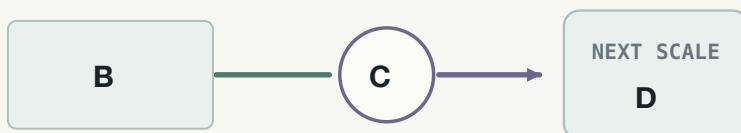
A Theory of Useful Differences

Form begins when differences hold one another without erasing their difference. The resulting whole can be wounded, repaired, and encountered as a new difference at the next scale.

ONE RECURSIVE MOVE



THE WHOLE ENTERS A NEW CONTACT



The theory does not begin with self-contained objects. It begins with oriented differences, their closure, and the future consequences of opening that closure.

THE AXIOM BEGINS BEFORE OBJECT, SCALE, AND OBSERVER

A body is a temporary balance of mutually sustaining differences, not a pre-given unity.

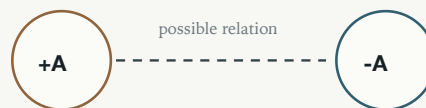
1 Difference. +A and -A are two orientations of one difference. They are not independent substances carrying intrinsic meaning before relation.

2 Contact. A meeting becomes productive when the two sides modify or complete one another and their joint consequence survives into a later contact.

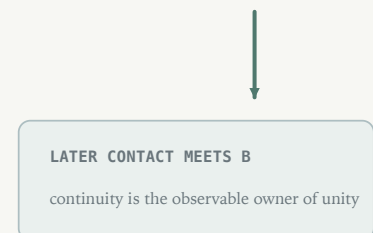
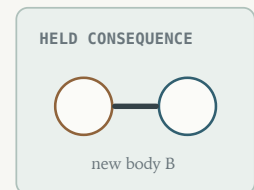
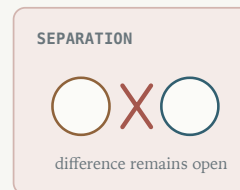
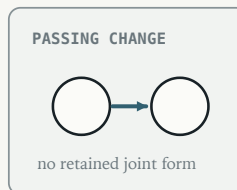
3 Body. The held relation is encountered as one operative unit while its internal difference remains causally available. Stability is renewed balance, not immobility.

THE SAME PAIR CAN PASS, SEPARATE, OR CLOSE INTO A CONTINUING BODY

BEFORE CONTACT · NO READY WHOLE



CONTACT HAS THREE PHYSICAL OUTCOMES



AXIOM

Balance does not cancel difference. It gives difference a body capable of continuing.

Axiom: mutually sustaining differences -> held relation -> one continuing body

SCALE IS PRODUCED BY CLOSURE RATHER THAN ASSIGNED FROM OUTSIDE

Two held sides continue as one new difference without losing their internal distinction.

1 One relation. An oriented relation has two distinguishable ends. This is the primitive 1:2: one whole participates through two sides.

2 Closure. Several relations may close into a surface or volume. Their lower causes remain inside; the closure gains one exterior interface.

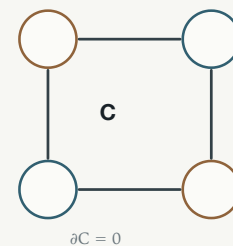
3 Recursion. The completed whole becomes a difference in another relation. No final scale or fixed number of internal parts is assumed.

RELATION → CLOSED CIRCUIT → VOLUME → ONE SIDE OF ANOTHER RELATION

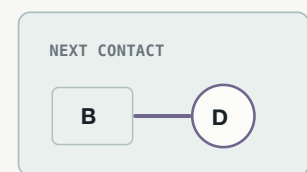
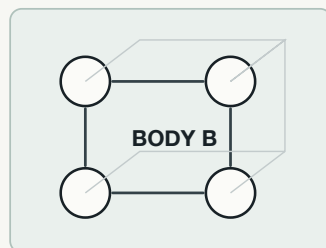
ONE ORIENTED RELATION



CLOSED SURFACE



THE CLOSED BODY GAINS AN EXTERIOR ROLE



internal differences remain routeable

SCALE TRANSITION

The whole is not a summary of its parts. It is their retained ability to meet the future as one.

$\partial C = 0$ DOES NOT MEAN $C = 0$

A wound opens the body's boundary; repair restores a causal role, not necessarily the missing piece.

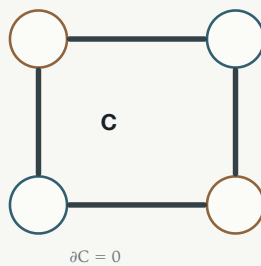
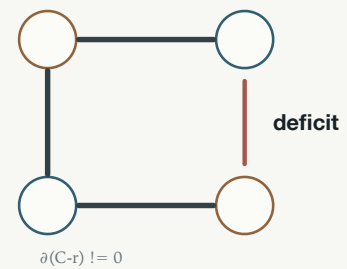
1 Closure. Let ∂ be the boundary operator. An oriented integral chain C is closed when $\partial C = 0$; its internal differences remain present.

2 Wound. Removing a necessary relation r gives $C - r$ an exposed boundary. The deficit changes the value of relations throughout the connected body.

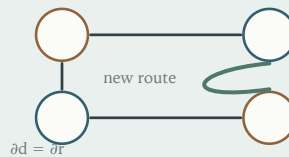
3 Repair. A completion d succeeds when $\partial d = \partial r$. It may restore, transform, divide, or stabilize another form.

CLOSED BODY $\rightarrow$ OPENED DEFICIT $\rightarrow$ TWO ADMISSIBLE FUTURES

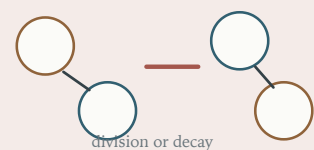
CLOSED BODY

REMOVE r 

THE DEFICIT CHANGES THE WHOLE FUTURE

COMPLETION d 

NO ADMISSIBLE FILLING



REPAIR THEOREM

Closure is recovered by boundary equivalence.
Identity of material is not required.

$$C' = C - r + d \cdot \text{closure restored iff } \partial d = \partial r$$

REGENERATION IS RETAINED CONSTRAINT, NOT REPLAY OF A FINISHED TARGET

An organism continues itself by progressively narrowing what can still complete it.

1 Open place. The deficit $D_t = \partial C_t$ is an unfinished relation, not a missing object with one prescribed replacement.

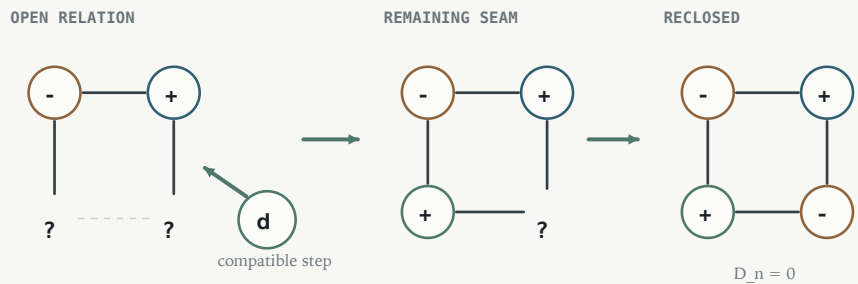
2 Possible continuations. $\Omega(C_t)$ contains every continuation that can still close from the body's retained distinctions.

3 Retention. A compatible d_t joins the same body. Continuations inconsistent with that earned relation are no longer available.

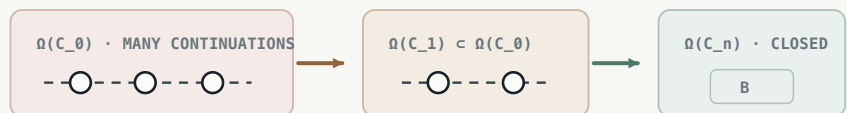
4 Recursion. When the remaining boundary reaches zero, the closure acts as one difference in a relation at the next scale.

EACH RETAINED COMPLETION NARROWS LIVE CONTINUATION UNTIL CLOSURE RECURS AT THE NEXT SCALE

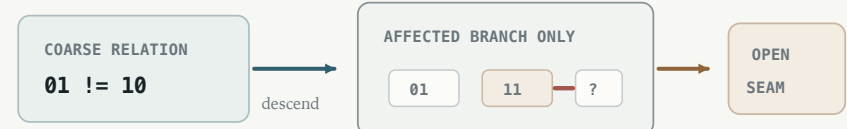
A BODY CAN COMPLETE ITSELF WITHOUT A STORED FINAL IMAGE



THE SET OF ADMISSIBLE CONTINUATIONS CONTRACTS



A MULTISCALE CARRIER CAN FEEL THE BREAK BEFORE LOCATING IT



REGENERATIVE SEARCH

The organism need not know its final form. It must retain steps that make its next closure more determinate.

The mmap hierarchy shown here is a proposed carrier architecture, not a result of the present assay.

$$D_t = \partial C_t \cdot \Omega(C_{t+1}) \subseteq \Omega(C_t) \cdot D_n = 0 \rightarrow \text{one new difference at the next scale}$$

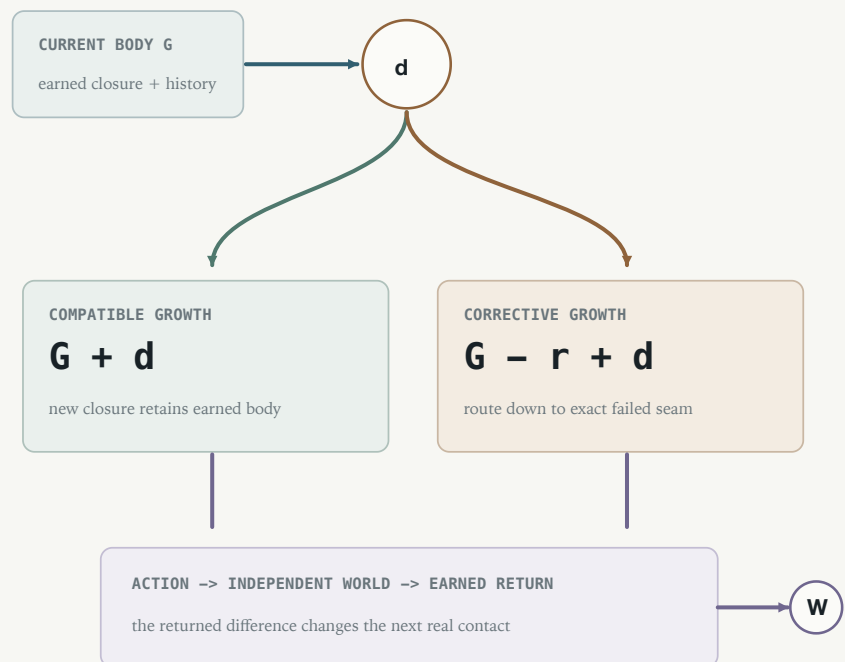
USEFULNESS IS EARNED THROUGH RETURN, NOT ASSIGNED BY A LABEL

A difference becomes useful only when its later consequences help the same body continue or revise itself.

1 Growth. A compatible difference d extends a live closure:
 $G_{next} = \text{close}(G+d)$. It becomes part of the body's next encounter.

2 Revision. If d exposes an earned relation r as false, the body performs local $G - r + d$ and preserves everything not causally implicated.

3 Return. The world must answer the changed action. A downstream witness compares live, blocked, and shuffled continuations; it does not choose the internal repair.

ONE CONTACT OPENS TWO LEGITIMATE PATHS: $+d$ AND $-r+d$ THE SAME d HAS NO CONTEXT-FREE VALUE

USEFUL DIFFERENCE

The future decides whether a difference was support, correction, or both.

`useful(d | G, contact, future) -> +d or -r+d -> world return -> changed continuation`

ELECTRICITY IS THE DYNAMICS OF OPEN DIFFERENCES

If charge is the oriented local homology class of a deficit, integral quantization follows from closure.

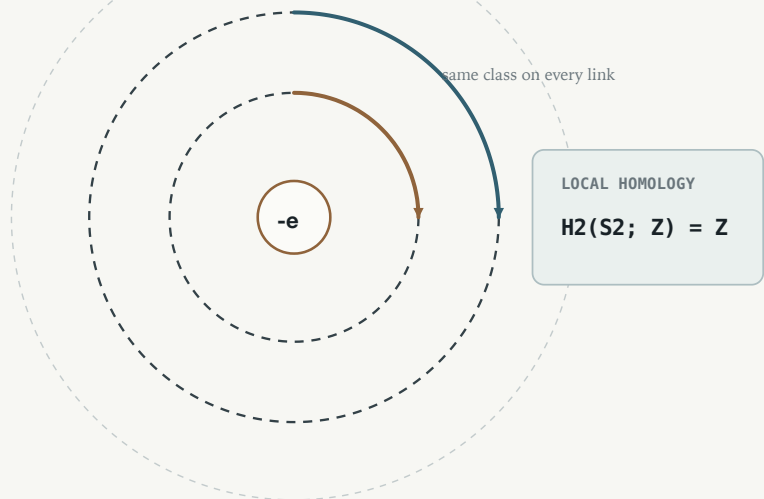
1 Deficit. Electric tension is an exposed difference propagated through the connected body. External balance means only $\partial C = \emptyset$ at that scale.

2 Local class. A point deficit in an oriented three-dimensional body has a surrounding link S^2 with $H_2(S^2; \mathbb{Z}) = \mathbb{Z}$.

3 Quantum. If the oriented generator g maps to $-e$, every allowed class is $n \cdot g$ and every charge is $-n \cdot e$.

EVERY SURROUNDING SURFACE READS THE SAME ORIENTED GENERATOR

LOCAL DEFICIT · NONLOCAL CLASS



INTEGER SECTORS



TOPOLOGICAL CONSEQUENCE

Charge cannot vary continuously inside an integral class. *Its sign is orientation; its multiplicity is an integer.*

The numerical scale e and the dynamical field law remain to be derived.

$$H_2(S^2; \mathbb{Z}) = \mathbb{Z} \cdot q(g) = -e \cdot q(n \cdot g) = -n \cdot e$$

OBSERVER PARTICIPATES · WITNESS READS THE SURVIVING CONSEQUENCE

A detector does not passively reveal a completed particle; it supplies a physical distinction that closes into a local trace.

1 Extended membership.

An electron is treated as a concentration of differences whose causal membership may extend beyond the eventual visible hit.

2 Observer. An

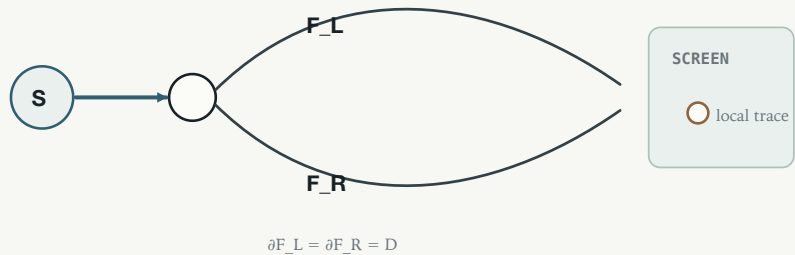
interferometer and path detector enter the event. They alter which alternatives are physically distinguishable and which closure can form.

3 Witness. A later reader

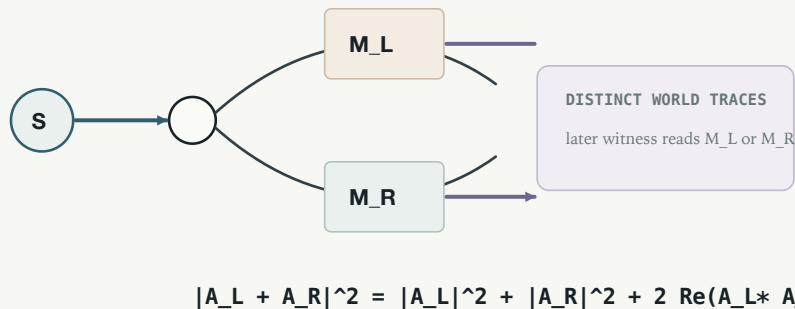
reconstructs the event from its stabilized trace. Reading is a new event, not a cause of the earlier result.

TWO UNRESOLVED FILLINGS SHARE ONE BOUNDARY; A PATH MARKER CREATES TWO DIFFERENT TRACES

UNRESOLVED COMPLETIONS



PATH DETECTOR SUPPLIES THE MISSING DISTINCTION



OPEN DERIVATION

No classical wave or corpuscle is required as the primitive. What remains is to derive complex filling weights and the interference term from closure.

THE SCALES SHARE A LAW, NOT A READY-MADE MECHANISM

One transition form connects physical closure, regeneration, cognition, and the persistent carrier.

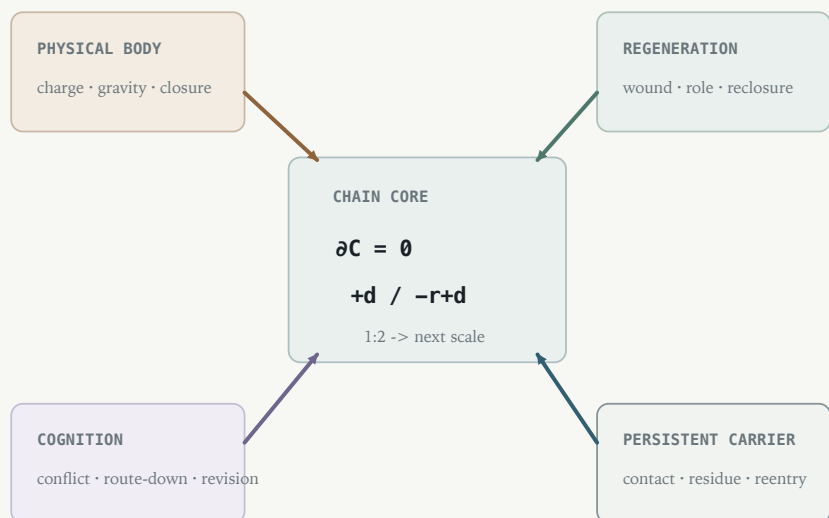
1 Invariant. Differences close, act as one, reopen under injury, and continue through $+d$ or $-r+d$. The transition is the common object.

2 Material owners. Physics, biology, cognition, and mmap must each provide their own carrier, contact, deficit, completion, and surviving consequence.

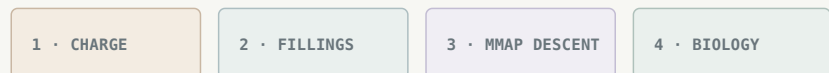
3 Decisive seam. The theory wins only where it derives a result or distinguishes causal-role completion from target replay, hidden controller, or passive storage.

COMMON ALGEBRA AT THE CENTER; DIFFERENT MATERIAL ASSAYS AT THE BOUNDARY

ONE FORM · FOUR MATERIAL SURFACES



NEXT DERIVATIONS



WHOLE

A form is the history of differences that can still repair and continue one another. Useful difference is the unit of its growth and self-revision.